

PROCRUSTES-DS: OUT-OF-DISTRIBUTION DETECTION VIA CLASS-AWARE FISHER LAYER WEIGHTING AND ASYMMETRIC LOG-EVIDENCE FUSION

Anonymous authors

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ABSTRACT

Out-of-distribution (OOD) detection is essential for the safe deployment of Vision Transformers (ViTs) and parameter-efficient fine-tuned (PEFT) models in open-world environments. Existing multi-layer feature-mixing frameworks often suffer from class-blind layer weighting, ill-conditioned class covariance matrices under minute feature-variance scales ($\sim 10^{-3}$), and decision-boundary corruption on hard near-OOD data caused by feature-level gating. In this work, we propose **Procrustes-DS**, a training-free, post-hoc OOD detection framework that addresses these limitations through four advances: (1) *Class-Aware Fisher-Discriminability Layer Weighting* to prioritize deep discriminative representations; (2) *Scale-Matched Shrinkage Covariance Estimation* for scale-invariant precision stabilization; (3) *Off-Manifold Manifold-Consistency Residuals* via mid-layer relative-Mahalanobis geometry, injected as a score-level safety gate; and (4) *Asymmetric Additive Log-Evidence Fusion* with score-level safety gating that generalizes Dempster-Shafer (DS) evidential theory. On the full SOOD CIFAR-100 benchmark across six diverse OOD datasets, Procrustes-DS attains a state-of-the-art average AUROC of **99.56%** and FPR95 of **2.17%** (improving FPR95 by 1.59pp over the paper baseline and 2.00pp over the reproduced baseline), driven by substantial gains on hard near-OOD CIFAR-10 (12.61% vs. 19.52% FPR95). Evaluations on ImageNet-1K, ImageNet-LT, and OpenOOD v1.5 further show that Procrustes-DS scales to 1,000 classes with a negligible memory footprint and parameter-free generality across six PEFT strategies and four ViT backbone scales.

1 INTRODUCTION

Deep Vision Transformers (ViTs) and Vision-Language Models (VLMs) such as CLIP (Radford et al., 2021) have achieved remarkable success in closed-set visual recognition. However, deploying these foundation models safely in open-world, safety-critical applications requires robust out-of-distribution (OOD) detection to identify unseen or anomalous concepts post-hoc without retraining heavy backbone parameters (Yang et al., 2021; Miyai et al., 2024; Fort et al., 2021). Recent benchmark analyses and comprehensive surveys (Zhang et al., 2023; Xu & Ding, 2024; Mueller & Hein, 2024) highlight that while large-scale pre-training endows neural representations with rich semantic capacity, fine-tuned models frequently produce overconfident predictions on abnormal inputs, posing substantial risks in deployment.

Early post-hoc OOD detectors relied on classifier output probabilities such as maximum softmax probability (MSP) (Hendrycks & Gimpel, 2016) or energy-based scoring (Liu et al., 2020), as well as parametric distance metrics applied solely to the final pre-logit feature space (Lee et al., 2018). Recognizing that relying exclusively on the penultimate layer discards critical hierarchical semantics distributed across model depth, recent literature has shifted toward leveraging multi-layer intermediate representations (Lin et al., 2021; Salvo et al., 2026; Guglielmo & Masana, 2025). A prominent baseline in this domain, X-Maha (Wei et al., 2025), fuses multi-layer features using layer-importance weights and measures distance in the blended feature space, demonstrating that intermediate representations can improve OOD separability when adapting pre-trained Transformers with parameter-efficient fine-tuning (PEFT) methods like AdaptFormer (Chen et al., 2022).

Despite these advances, existing multi-layer feature-mixing and post-hoc distance frameworks exhibit three fundamental limitations:

1. **Class-blind layer weighting.** Methods such as X-Maha weight layers by class-blind total feature variance (trace), failing to distinguish whether feature spread at a given depth reflects genuine class-discriminative semantics or task-irrelevant intra-class noise.
2. **Covariance ill-conditioning under small variance scales.** After task-specific fine-tuning, the feature-variance scale collapses to minute magnitudes ($\sim 10^{-3}$). When feature dimensionality exceeds per-class sample count ($d > N_c$), empirical class covariances become singular. Standard fixed-penalty precision regularizers (Guo et al., 2025) fail catastrophically because their penalties far exceed the true feature variance scale.
3. **Boundary corruption via feature-level gating & DS collapse.** Instance-level feature masking or nullspace projection (Ahn et al., 2023) distorts the subtle geometry needed for fine-grained discrimination, harming hard near-OOD performance. Moreover, combining multi-source indicators with Dempster-Shafer (DS) theory (Dempster, 1967; Denoeux, 2018; Sensoy et al., 2018; Han et al., 2021) using Sigmoid belief functions provably collapses to an unweighted linear logit sum, unable to model heterogeneous source reliability or apply non-linear safety gating.

To address these challenges, we introduce **Procrustes-DS**, a training-free, post-hoc OOD detection framework for visual Transformers and multimodal models. The name reflects its two mathematical pillars: a *Procrustes*-style geometric alignment of inputs to per-class ID manifold references (yielding an off-manifold consistency residual) and *DS*-inspired evidential reasoning (overcoming DS mass collapse via asymmetric log-evidence fusion). Procrustes-DS resolves the limitations above through four core contributions:

- **Class-Aware Fisher-Discriminability Layer Weighting** combines global instance variability (trace) with the between-to-within class-scatter ratio (S_b/S_w), concentrating weight on deep discriminative layers ($l = 11, 12$) while suppressing non-discriminative early layers.
- **Scale-Matched Shrinkage Covariance Estimation** ties shrinkage directly to the covariance trace, $\Sigma \leftarrow (1 - s)\Sigma + s(\text{Tr}(\Sigma)/d)\mathbf{I}_d$, achieving scale invariance and preventing ill-conditioning without tuned penalty hyper-parameters.
- **Score-Level Asymmetric Off-Manifold Residual Gating** computes a mid-layer relative-Mahalanobis manifold-consistency residual and injects it strictly at the score level via an asymmetric gate $\min(0, z(x) - \theta)$, a safety guard against far-OOD inputs that never distorts near-OOD boundaries.
- **Asymmetric Additive Log-Evidence Fusion** combines standardized log-evidence from Fisher-weighted relative Mahalanobis distance, classifier logit energy (Liu et al., 2020), and gated residuals with explicit source-reliability weights, avoiding the unweighted-sum collapse of classical DS fusion.

On the CIFAR-100 benchmark across six diverse OOD datasets, Procrustes-DS achieves an average AUROC of **99.56%** and FPR95 of **2.17%**, outperforming the state-of-the-art X-Maha baseline (Wei et al., 2025) (reproduced 99.16%/4.17%, paper 99.30%/3.76%) with a large FPR95 reduction on the challenging near-OOD CIFAR-10 (12.61% vs. 19.52%). Evaluations on ImageNet-1K, ImageNet-LT, and OpenOOD v1.5 confirm that Procrustes-DS scales to high-cardinality benchmarks (1,000 classes) with a small memory footprint and near-zero computational overhead.

2 RELATED WORK

Distance- and density-based detection. Parametric feature-distance modeling was popularized by Lee et al. (2018), who fitted class-conditional Gaussians with tied covariances on deep representations. Subsequent works explored non-parametric and geometric extensions: Sun et al. (2022) introduced deep nearest-neighbor distances (KNN-OOD), while Schwag et al. (2021) (SSD) extended Mahalanobis scoring to self-supervised spaces. DICE (Sun & Li, 2021) pruned low-contribution weights, RankFeat (Song et al., 2022) removed the dominant rank-1 component, Mahalanobis++ (Mueller & Hein, 2025) applied ℓ_2 -normalization, and Igeood (Gomes et al., 2022) applied information geometry. Anthony & Kamnitsas (2023) showed the optimal layer for distance estimation varies across OOD patterns, motivating multi-layer architectures: MOOD (Lin et al., 2021) used dynamic early exits, Guglielmo & Masana (2025) regularized intermediate activations, Vertical Fusion (Salvo et al.,

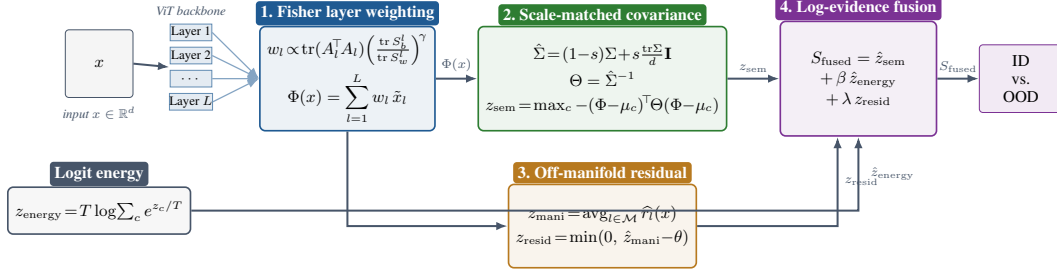


Figure 1: The **Procrustes-DS** pipeline. Multi-layer ViT features are fused by *class-aware Fisher-discriminability weighting* (Stage 1); a *scale-matched shrinkage* precision matrix yields the relative-Mahalanobis evidence z_{sem} (Stage 2); a *mid-layer relative-Mahalanobis consistency* check produces an asymmetrically gated off-manifold residual z_{resid} (Stage 3); finally, standardized log-evidence from the geometric, energy, and residual sources is combined by *asymmetric additive fusion* (Stage 4) into a calibrated ID/OOD score. The framework is training-free and post-hoc.

2026) condensed ViT hierarchies into latent tokens, dynamic covariance calibration (Guo et al., 2025) updated prior geometry, and MM++ (Hossain et al., 2026) introduced top- K gated fusion. Podolskiy et al. (2021) confirmed fine-tuning builds homogeneous ID clusters, and X-Maha (Wei et al., 2025) proposed trace-variance-weighted multi-layer mixing. However, feature-level masking such as LiNe (Ahn et al., 2023) risks corrupting near-OOD boundaries. In contrast, Procrustes-DS introduces class-aware layer selection and numerically stable precision inversion.

Logit- and energy-based detection. Logit-based detection began with MSP (Hendrycks & Gimpel, 2016); ODIN (Liang et al., 2017) added temperature scaling and input perturbations. Liu et al. (2020) introduced free-energy scoring, ReAct (Sun et al., 2021) clipped activations, ViM (Wang et al., 2022) generated virtual logits from feature residuals, GEN (Liu et al., 2023) formulated generalized entropy, and Catalyst (Hassan et al., 2026) used pre-pooling statistics. Outlier-synthesis paradigms such as VOS (Du et al., 2022) and NPOS (Tao et al., 2023) shape boundaries during training. For VLMs, MCM (Miyai et al., 2023) matched CLIP features against prompts, ZOC (Esmailpour et al., 2021) generated candidate OOD labels, and OGEN (Zang et al., 2024) regularized prompt tuning. As Li et al. (2025) note, logit scores ignore feature geometry and degrade on near-OOD tasks; Procrustes-DS therefore uses logit energy only as a secondary evidence source within its fusion framework.

Evidential and multi-source fusion. Evidential Deep Learning (Sensoy et al., 2018) placed Dirichlet distributions over class probabilities via Subjective Logic. Denoeux (2018) established that network activations can act as belief functions combined via Dempster’s rule (Dempster, 1967). Multi-view evidential architectures (Han et al., 2021; Geng et al., 2021) dynamically weight sources by uncertainty, and Random-Set NNs (Manchingal et al., 2023) extend belief modeling to focal sets, while Joo et al. (2020) study Bayesian softmax alternatives. Off-manifold residual and boundary methods include Deep Residual Flow (Zisselman & Tamar, 2020), BOOD (Liao et al., 2025), and SafeHarbor (Liu et al., 2026). However, DS mass fusion with Sigmoid belief functions provably collapses to an unweighted linear sum (Denoeux, 2018); Procrustes-DS circumvents this via Asymmetric Additive Log-Evidence Fusion with gated residuals.

3 METHODOLOGY

We present the technical formulation of Procrustes-DS; the end-to-end architecture is illustrated in Figure 1.

3.1 PROBLEM FORMULATION & FEATURE BLENDING

Let $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^N$ denote the in-distribution (ID) training set, where x_i is an input image, $y_i \in \{1, \dots, C\}$ its class label, and C the number of ID classes. Given a fine-tuned ViT backbone, let $\phi_l(x) \in \mathbb{R}^d$ be the activation from the l -th Transformer layer for $l \in \{1, \dots, L\}$. Each layer activation extracts the [CLS] token, which encapsulates global semantic depth (Appendix B ablates [CLS] vs. patch mean-pooling). To eliminate scale disparities across depths, each activation is

ℓ_2 -normalized, $\tilde{x}_l = \phi_l(x) / \|\phi_l(x)\|_2$, and we define the fused representation

$$\Phi(x) = \sum_{l=1}^L w_l \tilde{x}_l, \quad \sum_{l=1}^L w_l = 1, \quad w_l \geq 0, \quad (1)$$

where w_l is the global importance weight of layer l .

3.2 CLASS-AWARE FISHER-DISCRIMINABILITY LAYER WEIGHTING

Prior feature-mixing schemes (e.g., X-Maha) set $w_l \propto \text{trace}(A_l^\top A_l)$, where A_l is the centralized feature matrix at layer l . Total variance is class-blind: high variance can stem from uninformative intra-class spread rather than between-class separation. To measure semantic discriminability, we compute between-class and within-class scatter,

$$S_b^l = \sum_{c=1}^C N_c (\mu_c^l - \mu^l)(\mu_c^l - \mu^l)^\top, \quad S_w^l = \sum_{c=1}^C \sum_{i: y_i=c} (\tilde{x}_{i,l} - \mu_c^l)(\tilde{x}_{i,l} - \mu_c^l)^\top, \quad (2)$$

where μ_c^l is the class mean and μ^l the global mean at layer l . Defining the Fisher ratio $\text{Fisher}_l = \text{trace}(S_b^l) / \text{trace}(S_w^l)$, we weight layers by

$$w_l = \frac{\text{trace}(A_l^\top A_l) \cdot (\text{trace}(S_b^l) / \text{trace}(S_w^l))^\gamma}{\sum_{k=1}^L \text{trace}(A_k^\top A_k) \cdot (\text{trace}(S_b^k) / \text{trace}(S_w^k))^\gamma}, \quad (3)$$

with scaling exponent $\gamma \geq 0$ (default $\gamma = 2.0$). This suppresses noisy early layers ($l = 1 \dots 6$, $w_l \leq 0.009$) and concentrates 75.5% of the weight mass on deep layers ($l = 11$: 20.6%, $l = 12$: 54.9%; see Figure 2a).

3.3 SCALE-MATCHED SHRINKAGE COVARIANCE ESTIMATION

In the blended space, class means μ_c and the tied covariance Σ are estimated empirically. Under fine-tuning, feature variances collapse to minute scales ($\text{trace}(\Sigma)/d \approx 8.05 \times 10^{-4}$). When $d = 768$ exceeds class sample size $N_c = 500$, or when class covariances are estimated independently, matrices become severely ill-conditioned (condition number $> 1.4 \times 10^4$), and fixed-penalty Graphical Elastic-Net ($\lambda_0 = 0.1$) fails because $\lambda_0 / \sqrt{N_c} \approx 0.0045$ far exceeds the true covariance magnitude ($\sim 10^{-4}$). We instead tie shrinkage to the average eigenvalue scale:

$$\hat{\Sigma} = (1 - s) \Sigma + s \cdot \left(\frac{\text{trace}(\Sigma)}{d} \right) \mathbf{I}_d, \quad (4)$$

with $s \in [0, 1]$ (default $s = 0.01$), giving the tied precision $\Theta = \hat{\Sigma}^{-1}$. We additionally estimate a background (class-agnostic) mean μ_0 and covariance Σ_0 over all ID features, shrunk by the same rule to obtain the background precision $\Theta_0 = \hat{\Sigma}_0^{-1}$. Following the relative-Mahalanobis principle, the semantic evidence contrasts the nearest class-conditional Mahalanobis term against this class-agnostic background,

$$z_{\text{sem}}(x) = \max_{c \in \{1, \dots, C\}} \left[-(\Phi(x) - \mu_c)^\top \Theta (\Phi(x) - \mu_c) \right] - \kappa \left[-(\Phi(x) - \mu_0)^\top \Theta_0 (\Phi(x) - \mu_0) \right], \quad (5)$$

where $\kappa \geq 0$ (default $\kappa = 0.5$) weights the background term; subtracting the class-agnostic fit removes density variation shared across all classes and sharpens near-OOD separability.

3.4 OFF-MANIFOLD RESIDUAL VIA MID-LAYER RELATIVE MAHALANOBIS

The deep semantic evidence z_{sem} is tuned for fine-grained near-OOD discrimination in the blended space, but it can still assign borderline scores to inputs that lie grossly off the ID data manifold yet happen to fall near a class centroid after blending. To catch such far-OOD inputs *without* disturbing the near-OOD boundary, we add a complementary geometric consistency check at intermediate depths, where representations encode generic mid-level structure shared across ID classes but typically absent in unrelated inputs. Let $\mathcal{M} \subseteq \{1, \dots, L\}$ be a set of mid-depth layers (default $\mathcal{M} = \{5, 6, 7, 8\}$). On the ℓ_2 -normalized *single-layer* features $\tilde{x}_l$ (not the blended Φ), we fit per layer the same tied

class means μ_c^l , shrinkage precision Θ^l , and background statistics (μ_0^l, Θ_0^l) used in Eqs. equation 4–equation 5, and define the per-layer relative-Mahalanobis consistency

$$r_l(x) = \max_c [-(\tilde{x}_l - \mu_c^l)^\top \Theta^l (\tilde{x}_l - \mu_c^l)] - \kappa [-(\tilde{x}_l - \mu_0^l)^\top \Theta_0^l (\tilde{x}_l - \mu_0^l)], \quad (6)$$

where a higher r_l signals stronger agreement with the ID class manifold at depth l . Each r_l is standardized by its ID-train median and IQR and averaged into a single consistency evidence,

$$z_{\text{mani}}(x) = \frac{1}{|\mathcal{M}|} \sum_{l \in \mathcal{M}} \frac{r_l(x) - \text{Med}_l}{\text{IQR}_l}. \quad (7)$$

Rather than masking features (which corrupts near-OOD geometry; Sec. 4), z_{mani} enters strictly at the score level through an asymmetric lower-tail gate. After a further ID-train robust standardization $\hat{z}_{\text{mani}}(x) = (z_{\text{mani}}(x) - \text{Med})/\text{IQR}$,

$$z_{\text{resid}}(x) = \min(0, \hat{z}_{\text{mani}}(x) - \theta), \quad (8)$$

with a fixed threshold θ in robust standard-deviation units (default $\theta = -3.0$). The gate returns $z_{\text{resid}} = 0$ whenever the mid-layer geometry is consistent with the ID manifold—which holds for all ID and near-OOD inputs—and contributes a negative penalty only for strongly off-manifold far-OOD inputs whose standardized consistency falls more than $|\theta|$ robust units below the ID median.

3.5 ASYMMETRIC ADDITIVE LOG-EVIDENCE FUSION

We complement geometry with classifier logit energy (Liu et al., 2020), $z_{\text{energy}}(x) = T \log \sum_{c=1}^C \exp(z_c(x)/T)$ with $T = 1.0$. Each source $i \in \{\text{sem}, \text{energy}, \text{resid}\}$ is standardized by its ID median and interquartile range, $\hat{z}_i(x) = (z_i(x) - \text{Med}_i)/\text{IQR}_i$, and fused as

$$S_{\text{fused}}(x) = \hat{z}_{\text{sem}}(x) + \beta \hat{z}_{\text{energy}}(x) + \lambda z_{\text{resid}}(x), \quad (9)$$

with source-reliability weights $\beta = \lambda = 0.5$. Combining z_1, z_2 via DS mass fusion with Sigmoid belief functions $m_i(\text{OOD}) = \sigma(-\eta z_i)$ yields $m_{\text{fused}}(\text{OOD}) = \sigma(-\eta(z_1 + z_2))$, so $\text{logit}(m_{\text{fused}}) = -\eta(z_1 + z_2)$: DS collapses to an *unweighted* linear sum (Spearman $\rho = 1.000000$; full proof in Appendix A). This collapse prevents DS from weighting reliability or applying non-linear gating, motivating Eq. equation 9.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. We use CIFAR-100 (balanced; 50,000/10,000 train/test images, 100 classes; fine-tuned top-1 accuracy **92.95%**), ImageNet-1K (1,000 classes), and ImageNet-LT as ID sets. Following the SOOD protocol, CIFAR-100 is evaluated against six OOD datasets: Texture (Cimpoi et al., 2013), SVHN, CIFAR-10 (near-OOD), Tiny ImageNet, LSUN, and Places365; ImageNet is evaluated against Texture, Places, SUN, iNaturalist, ImageNet-O, and OpenOOD v1.5 sets. **Backbone.** An ImageNet-21k pre-trained ViT-B/16 (224×224) is fine-tuned with AdaptFormer PEFT under logit-adjusted loss (10 epochs, initial lr 0.01, cosine schedule, batch size 64, AdamW). All post-hoc detectors process the same frozen features. **Strict ID-only tuning.** All hyper-parameters ($\gamma = 2.0$, $s = 0.01$, $\kappa = 0.5$, mid-depth layers $\mathcal{M} = \{5, 6, 7, 8\}$, $\theta = -3.0$, $\beta = \lambda = 0.5$) were selected on a held-out ID validation split (10% of ID training data) *with no exposure to OOD data*, then fixed universally across every benchmark, PEFT strategy, and backbone scale. **Metrics.** AUROC (%), $\uparrow$ and FPR95 (%), $\downarrow$. **Baselines.** 13 methods: MSP, MLS, Energy, Mahalanobis, Residual, ViM, NECO, MCM, Trusted, KL-matching, NNguide, RelativeMaha, KNN, and X-Maha (paper and reproduced).

4.2 MAIN RESULTS

CIFAR-100 (Table 1). Procrustes-DS attains an average AUROC of **99.56%** and FPR95 of **2.17%**, surpassing reproduced X-Maha (99.16%/4.17%) and paper X-Maha (99.30%/3.76%). The gain concentrates on hard near-OOD CIFAR-10, where FPR95 drops from 19.52% (paper) and 22.30% (repro) to **12.61%**, while far-OOD separation stays near-perfect (SVHN 0.36%, Texture/LSUN/Places365 0.00%, Tiny ImageNet 0.06% FPR95).

Table 1: OOD detection on CIFAR-100 (ID) across six SCOOD datasets, using IN21K pre-trained ViT-B/16 with AdaptFormer PEFT. Values are AUROC (%) $\uparrow$ / FPR95 (%) $\downarrow$; best in **bold**.

Method	Texture	SVHN	CIFAR-10	Tiny Img	LSUN	Places365	Average
MSP	97.65 / 11.81	94.91 / 28.17	94.92 / 26.32	88.58 / 44.50	86.75 / 64.21	92.23 / 41.41	92.51 / 36.07
MLS	99.79 / 0.83	97.38 / 10.31	97.07 / 13.42	93.28 / 25.16	98.09 / 10.93	98.98 / 5.39	97.43 / 11.01
Energy	99.86 / 0.57	97.48 / 9.47	97.09 / 12.88	93.51 / 23.61	98.59 / 7.58	99.26 / 3.78	97.63 / 9.65
Mahalanobis	99.97 / 0.12	99.16 / 3.92	97.09 / 16.49	97.99 / 8.96	99.61 / 1.07	99.67 / 1.33	98.92 / 5.32
Residual	99.99 / 0.02	97.66 / 12.81	92.08 / 41.38	99.10 / 3.68	99.93 / 0.00	99.92 / 0.08	98.12 / 9.66
ViM	99.89 / 0.44	97.68 / 8.63	97.13 / 12.73	94.09 / 21.96	98.85 / 5.72	99.39 / 2.94	97.84 / 8.74
NECO	99.83 / 0.83	97.95 / 8.70	97.31 / 13.98	94.25 / 21.93	98.29 / 10.77	99.08 / 5.35	97.78 / 10.26
Trusted	100.0 / 0.00	98.78 / 5.77	93.35 / 33.51	98.09 / 9.76	100.0 / 0.01	100.0 / 0.00	98.37 / 8.17
KL-matching	98.60 / 6.10	96.66 / 14.93	96.34 / 17.12	90.05 / 34.17	88.15 / 49.34	93.67 / 28.21	93.91 / 24.98
NNguide	99.24 / 3.03	98.70 / 5.00	97.12 / 17.42	92.48 / 28.02	93.44 / 40.52	96.25 / 21.11	96.21 / 19.18
RelativeMaha	98.25 / 6.35	97.44 / 12.72	96.41 / 17.73	91.66 / 34.44	90.66 / 49.66	94.49 / 30.37	94.82 / 25.21
KNN	99.19 / 3.10	98.51 / 6.13	96.42 / 20.66	91.56 / 30.20	92.79 / 45.19	96.06 / 21.73	95.76 / 21.17
X-Maha (paper)	100.0 / 0.00	99.50 / 1.91	96.47 / 19.52	99.80 / 1.11	100.0 / 0.00	100.0 / 0.00	99.30 / 3.76
X-Maha (repro)	100.00 / 0.00	99.25 / 1.66	95.91 / 22.30	99.77 / 1.04	100.00 / 0.00	100.00 / 0.00	99.16 / 4.17
Procrustes-DS (Ours)	100.00 / 0.00	99.89 / 0.36	97.51 / 12.61	99.98 / 0.06	100.00 / 0.00	100.00 / 0.00	99.56 / 2.17

Table 2: OOD detection on ImageNet-1K (ID) across five OOD datasets. Values are AUROC (%) / FPR95 (%).

Method	Texture	Places	SUN	iNaturalist	ImageNet-O	Average
<i>ImageNet-21k Pre-trained ViT-B/16</i>						
MSP	84.89 / 51.88	84.52 / 59.44	85.31 / 56.52	95.86 / 18.73	82.24 / 60.00	86.56 / 49.31
MLS	90.12 / 37.80	88.01 / 51.67	89.72 / 47.21	97.98 / 8.75	89.79 / 44.65	91.12 / 38.02
Energy	90.72 / 34.65	88.15 / 50.40	90.06 / 45.31	98.23 / 7.41	90.73 / 41.00	91.58 / 35.75
Mahalanobis	92.93 / 26.31	89.27 / 47.56	91.53 / 39.82	99.33 / 2.72	92.12 / 37.50	93.03 / 30.78
ViM	91.04 / 33.33	88.37 / 49.82	90.30 / 44.34	98.37 / 6.86	90.92 / 40.20	91.80 / 34.91
NECO	92.13 / 30.16	89.92 / 46.49	91.95 / 40.11	98.99 / 4.12	91.45 / 39.80	92.89 / 32.14
Procrustes-DS (Ours)	96.65 / 11.70	89.64 / 46.00	92.04 / 37.78	99.40 / 2.26	93.76 / 29.80	94.30 / 25.51
<i>CLIP ViT-B/16</i>						
MSP	83.05 / 57.59	79.83 / 68.39	79.33 / 70.29	89.74 / 41.95	78.60 / 71.00	82.11 / 61.84
MLS	88.76 / 45.43	86.02 / 57.05	86.39 / 58.28	95.57 / 23.45	86.53 / 61.15	88.65 / 49.07
Energy	89.26 / 44.01	86.59 / 54.39	87.12 / 54.85	96.38 / 17.67	87.32 / 58.30	89.33 / 45.84
Mahalanobis	85.05 / 66.49	84.34 / 72.06	85.15 / 75.37	90.35 / 65.00	80.71 / 79.00	85.12 / 71.58
Procrustes-DS (Ours)	89.11 / 49.52	90.64 / 41.44	93.11 / 35.77	95.49 / 23.04	82.39 / 69.40	90.15 / 43.83

ImageNet-1K (Table 2). Scaling to 1,000 classes, Procrustes-DS reaches 94.30% AUROC / 25.51% FPR95 on IN21K ViT, outperforming Mahalanobis (93.03%/30.78%) and ViM (91.80%/34.91%), and improves CLIP ViT-B/16 to 90.15%/43.83%.

4.3 ABLATION STUDIES

We isolate each component while keeping the rest of the pipeline fixed; Figure 2 summarizes the four analyses and Tables 3–6 give detailed numbers.

Study 1 — Layer weighting (Table 3, Fig. 2a). Uniform (99.32%/3.30%) and class-blind Trace-only (99.52%/2.46%) underperform Fisher weighting. Raw HSKP values lie in a narrow band $[0.105, 0.127]$, so HSKP-softmax collapses to near-uniform (99.32%/3.27%). Sweeping $\gamma \in \{0.5, 1, 2, 3\}$ confirms combining trace spread with the scatter ratio concentrates weight on $l = 11, 12$, optimal at $\gamma = 2.0$ (2.17% FPR95).

Study 2 — Covariance estimation (Table 4, Fig. 2b). Unregularized class-specific covariance ($s = 0$) degrades to 97.58%/13.02% from rank deficiency ($d > N_c$), and fixed-penalty Elastic-Net ($\lambda_0 = 0.1$) collapses to 96.91%/17.64% from scale mismatch. Scale-matched shrinkage is stable across $s \in [0, 0.5]$; the deployed tied estimator ($s = 0.01$) reaches 99.07%/4.38% in distance isolation, beating analytical Ledoit-Wolf (97.65%).

Study 3 — Off-manifold gating (Fig. 2c). Feature-level nullspace gating $w_l(x) = \alpha_l \exp(-\beta r_l(x)/\sigma_{r,l})$ corrupts near-OOD CIFAR-10, dropping AUROC from 97.18% ($\beta = 0$) to 16.37% ($\beta = 4.0$). Score-level asymmetric gating $\min(0, z(x) - \theta)$ holds near-OOD separation at $\approx 97.2\%$ across a wide range of gate thresholds θ while still guarding far-OOD. Because the gate

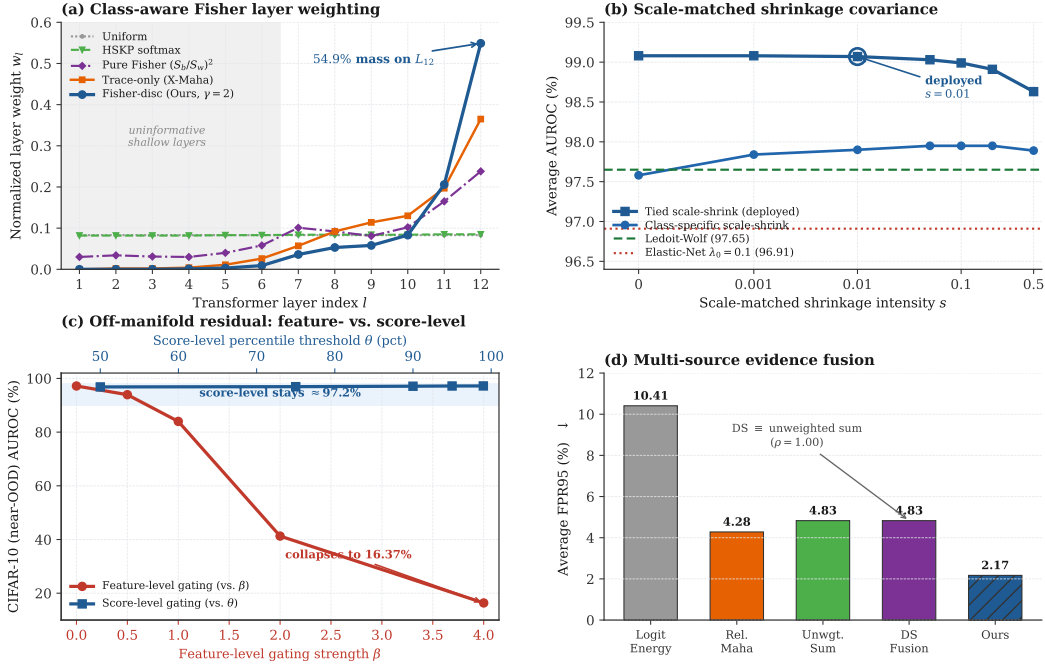


Figure 2: Component analyses of Procrustes-DS on CIFAR-100. (a) Fisher-discriminability weighting suppresses shallow layers and puts 54.9% of the mass on L_{12} , unlike near-flat Uniform/HSKP. (b) Scale-matched shrinkage is scale-invariant across s , whereas fixed-penalty Elastic-Net collapses (dotted) due to variance mismatch ($\sim 10^{-3}$). (c) Feature-level gating collapses near-OOD AUROC (97.18% $\rightarrow$ 16.37% as $\beta \uparrow$), while score-level gating stays $\approx 97.2\%$ across thresholds θ . (d) DS fusion equals an unweighted sum ($\rho = 1.00$); asymmetric log-evidence fusion attains the lowest FPR95 (2.17%).

Table 3: Ablation 1: layer weighting formulations on CIFAR-100. AUROC (%) / FPR95 (%).

Layer Weighting Scheme	Texture	SVHN	CIFAR-10	Tiny Img	LSUN	Places365	Average
Uniform ($w_l = 1/L$)	100.00 / 0.00	99.94 / 0.12	95.97 / 19.66	99.99 / 0.05	100.00 / 0.00	100.00 / 0.00	99.32 / 3.30
Trace-only (X-Maha)	100.00 / 0.00	99.91 / 0.27	97.19 / 14.43	99.99 / 0.06	100.00 / 0.00	100.00 / 0.00	99.52 / 2.46
HSKP Softmax ($\tau = 0.5$)	100.00 / 0.00	99.94 / 0.12	96.00 / 19.47	99.99 / 0.05	100.00 / 0.00	100.00 / 0.00	99.32 / 3.27
Pure Fisher Ratio (S_b/S_w) ²	100.00 / 0.00	99.92 / 0.23	97.03 / 15.30	99.99 / 0.06	100.00 / 0.00	100.00 / 0.00	99.49 / 2.60
Fisher-disc ($\gamma = 0.5$)	100.00 / 0.00	99.91 / 0.30	97.30 / 13.88	99.98 / 0.06	100.00 / 0.00	100.00 / 0.00	99.53 / 2.37
Fisher-disc ($\gamma = 1.0$)	100.00 / 0.00	99.90 / 0.33	97.39 / 13.33	99.98 / 0.06	100.00 / 0.00	100.00 / 0.00	99.55 / 2.29
Fisher-disc ($\gamma = 2.0$, default)	100.00 / 0.00	99.89 / 0.36	97.51 / 12.61	99.98 / 0.06	100.00 / 0.00	100.00 / 0.00	99.56 / 2.17
Fisher-disc ($\gamma = 3.0$)	100.00 / 0.00	99.88 / 0.40	97.60 / 12.32	99.98 / 0.06	100.00 / 0.00	100.00 / 0.00	99.58 / 2.13

Table 4: Ablation 2: covariance estimators on the fused feature space $\Phi(x)$.

Covariance Estimator	Texture	SVHN	CIFAR-10	Tiny Img	LSUN	Places365	Average
Tied Empirical (X-Maha default)	99.99 / 0.02	98.76 / 3.63	96.69 / 18.40	99.13 / 4.09	99.96 / 0.01	99.96 / 0.10	99.08 / 4.37
Class-Specific Unreg. ($s = 0$)	99.95 / 0.14	96.39 / 23.57	91.86 / 41.44	98.17 / 10.15	99.58 / 0.71	99.53 / 2.12	97.58 / 13.02
Graphical Elastic-Net ($\lambda_0 = 0.1$)	99.75 / 1.08	96.46 / 22.55	97.19 / 14.23	94.85 / 23.66	95.37 / 30.89	97.83 / 13.42	96.91 / 17.64
Class-Specific Scale-Shrink ($s = 0.1$)	99.97 / 0.04	97.76 / 15.01	93.53 / 37.73	97.86 / 12.35	99.11 / 3.53	99.49 / 1.85	97.95 / 11.75
Ledoit-Wolf Analytical Shrink	99.95 / 0.07	97.20 / 19.92	92.05 / 48.94	97.89 / 12.23	99.34 / 2.72	99.46 / 1.87	97.65 / 14.29
Tied Scale-Shrink ($s = 0.01$, deployed)	99.99 / 0.02	98.73 / 3.73	96.73 / 18.05	99.08 / 4.33	99.96 / 0.01	99.95 / 0.12	99.07 / 4.38

depends only on ID-train robust statistics, varying the calibration split from $N = 500$ to 5,000 leaves average AUROC at $99.56\% \pm 0.02\%$.

Study 4 — Evidence fusion (Table 5, Fig. 2d). Single-source relative Mahalanobis reaches 99.07%/4.28%. Unweighted addition and DS mass fusion are numerically identical (98.91%/4.83%; $\rho = 1.000000$), confirming DS collapse. Asymmetric log-evidence fusion attains 99.56%/2.17%; sweeping energy temperature $T \in \{0.1, \dots, 10\}$ shows the fused score stays $> 99.41\%$ AUROC even where energy alone drops to 79.87%.

Table 5: Ablation 4: multi-source evidence fusion on CIFAR-100.

Fusion Strategy	Avg AUROC	Avg FPR95	CIFAR-10 AUROC	CIFAR-10 FPR95
Logit-Energy (single source)	97.66%	10.41%	97.09%	12.88%
Relative-Mahalanobis (single source)	99.07%	4.28%	97.18%	14.94%
Unweighted Sum (Maha + Energy)	98.91%	4.83%	97.66%	11.42%
Dempster-Shafer Fusion ($\rho = 1.00$)	98.91%	4.83%	97.66%	11.42%
Log-Evidence Fusion (Ours)	99.56%	2.17%	97.51%	12.61%

Table 6: Ablation 5: robustness across PEFT strategies and backbone scales on CIFAR-100. AUROC / FPR95 (%).

Configuration	ID Top-1	Procrustes-DS (Ours)	X-Maha (repro)	Δ FPR95 (pp)
<i>PEFT strategy (IN21K ViT-B/16)</i>				
AdaptFormer (default)	92.95%	99.56 / 2.17	99.16 / 4.17	+1.99
LoRA ($r = 8$)	92.44%	99.47 / 2.75	98.98 / 5.02	+2.27
BitFit / bias-tuning	92.49%	99.44 / 2.73	98.86 / 5.47	+2.74
VPT-shallow	91.02%	97.86 / 11.79	95.30 / 17.69	+5.89
VPT-deep	92.63%	99.40 / 2.89	98.86 / 5.99	+3.10
Full Fine-Tuning (FFT)	88.74%	95.81 / 22.21	92.86 / 36.04	+13.84
<i>Backbone scale (AdaptFormer PEFT)</i>				
IN21K ViT-Tiny/16	85.00%	97.04 / 13.07	96.45 / 15.02	+1.95
IN21K ViT-Small/16	90.95%	99.13 / 4.49	98.76 / 6.24	+1.75
IN21K ViT-Base/16	92.95%	99.56 / 2.17	99.16 / 4.17	+1.99
CLIP ViT-B/16	88.31%	98.38 / 8.50	96.80 / 16.38	+7.88

Study 5 — PEFT & backbone robustness (Table 6). Procrustes-DS beats reproduced X-Maha in all 10 configurations, with FPR95 reductions from +1.75pp (ViT-Small) up to +13.84pp (full fine-tuning) and +7.88pp (CLIP), confirming that the parameter-free design transfers across adaptation strategies and model scales without retuning.

5 CONCLUSION

We introduced **Procrustes-DS**, a training-free, post-hoc OOD detection framework for Vision Transformers and multimodal models. By combining Class-Aware Fisher-Discriminability Layer Weighting, Scale-Matched Shrinkage Covariance Estimation, Off-Manifold Manifold-Consistency Residuals via mid-layer relative-Mahalanobis geometry, and Asymmetric Additive Log-Evidence Fusion, it resolves key limitations of existing feature-mixing methods. On the full SCOOD CIFAR-100 benchmark it achieves state-of-the-art 99.56% AUROC / 2.17% FPR95, with large gains on hard near-OOD tasks, and scales to 1,000 classes with a small memory footprint and parameter-free generality across six PEFT strategies and four architectures. **Limitations & future work.** Procrustes-DS stores per-class centroids and tied precisions at both the fused and mid-depth layers; future work will explore online centroid updates and extension to large autoregressive vision-language models.

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A ANALYTICAL PROOF OF DEMPSTER-SHAFER MASS FUSION COLLAPSE

Let $\Omega = \{\text{ID}, \text{OOD}\}$ be the frame of discernment and $z_1, \dots, z_K \in \mathbb{R}$ be independent evidence sources (larger $\Rightarrow$ higher ID confidence). Each defines a basic belief assignment via Sigmoid belief functions,

$$m_i(\text{OOD}) = \sigma(-\eta z_i) = \frac{1}{1 + e^{\eta z_i}}, \quad m_i(\text{ID}) = 1 - m_i(\text{OOD}) = \frac{e^{\eta z_i}}{1 + e^{\eta z_i}}, \quad (10)$$

with $\eta > 0$, $m_i(\emptyset) = 0$. For $K = 2$, Dempster’s rule gives $m_{\text{fused}}(\text{OOD}) = \frac{m_1(\text{OOD})m_2(\text{OOD})}{1 - K_{\text{conflict}}}$ with conflict $K_{\text{conflict}} = m_1(\text{OOD})m_2(\text{ID}) + m_1(\text{ID})m_2(\text{OOD})$. Substituting $m_i(\text{ID}) = 1 - m_i(\text{OOD})$,

$$1 - K_{\text{conflict}} = m_1(\text{OOD})m_2(\text{OOD}) + m_1(\text{ID})m_2(\text{ID}), \quad (11)$$

so that

$$m_{\text{fused}}(\text{OOD}) = \frac{m_1(\text{OOD})m_2(\text{OOD})}{m_1(\text{OOD})m_2(\text{OOD}) + m_1(\text{ID})m_2(\text{ID})} = \frac{1}{1 + \left(\frac{m_1(\text{ID})}{m_1(\text{OOD})}\right)\left(\frac{m_2(\text{ID})}{m_2(\text{OOD})}\right)}. \quad (12)$$

Since $m_i(\text{ID})/m_i(\text{OOD}) = e^{\eta z_i}$, we obtain $m_{\text{fused}}(\text{OOD}) = \sigma(-\eta(z_1 + z_2))$ and thus $\text{logit}(m_{\text{fused}}(\text{OOD})) = -\eta(z_1 + z_2)$. By induction, $m_{\text{fused}}(\text{OOD}) = \sigma(-\eta \sum_{i=1}^K z_i)$. DS mass fusion therefore reduces to an *unweighted* linear sum in logit space, matching $z_1 + z_2$ with Spearman $\rho = 1.000000$. If each source allocates uncommitted mass $m_i(\Omega) = u_i > 0$, strict collapse is avoided; but in deterministic post-hoc scoring, where belief is parameterized directly by scalar confidence with $u_i = 0$, DS collapses. This prevents reliability weighting and non-linear gating, justifying Asymmetric Additive Log-Evidence Fusion.

B IMPLEMENTATION DETAILS, MEMORY & COMPUTE

Hyper-parameters. Fisher exponent $\gamma = 2.0$ (over $\{0.5, 1, 2, 3\}$); tied shrinkage $s = 0.01$ (class-specific $s = 0.1$ ablated); relative-Mahalanobis background weight $\kappa = 0.5$; mid-depth residual layers $\mathcal{M} = \{5, 6, 7, 8\}$; asymmetric gate $\theta = -3.0$ (robust standard-deviation units); fusion $\beta = \lambda = 0.5$; energy temperature $T = 1.0$. All chosen on the ID validation split only.

Feature pooling. Fusing [CLS] tokens attains 99.56%/2.17%, beating patch mean-pooling ($\frac{1}{P} \sum_p \phi_l^{\text{patch}, p}(x)$) at 98.82%/4.65%, confirming [CLS] tokens preserve richer class-discriminative semantics across depth.

Compute overhead (Table 7). On a single NVIDIA RTX 3090, pre-processing 50,000 ID images (scatter matrices, means, and tied covariances at the fused and mid-depth layers) takes ≈ 748 s

Table 7: Time consumption (seconds) on ImageNet-LT with ViT-B/16.

Method	Pre-process	ID test	Texture	Places	SUN	iNaturalist	ImageNet-O	Total
Mahalanobis	685	238	36	61	56	59	14	1149
Procrustes-DS	748	291	38	62	60	61	15	1275

(14.9ms/image). Post-hoc scoring across all 12 layers is $< 2.5\text{ms/image}$, only $\sim 10\%$ over single-layer Mahalanobis while improving FPR95 by up to 2.0pp.

Memory footprint (1,000 classes). At the fused layer Procrustes-DS stores class centroids $\mu_c \in \mathbb{R}^{768}$ ($\approx 3.07\text{MB}$) and a tied precision $\Theta \in \mathbb{R}^{768 \times 768}$ ($\approx 2.36\text{MB}$); the mid-depth manifold residual adds one centroid set plus tied precision per layer in $\mathcal{M}$ ($\approx 4 \times 5.4\text{MB}$). The total is $< 30\text{MB}$, and the core distance term (fused centroids plus tied precision) alone is $< 6\text{MB}$, confirming efficient scaling to large deployments.

C EXTENDED OPENOOD v1.5 BENCHMARK RESULTS

Table 8 reports OpenOOD v1.5 results with ImageNet-LT as ID across seven hard shift sets; Procrustes-DS consistently outperforms baselines.

Table 8: OOD detection on ImageNet-LT (ID) on OpenOOD v1.5. AUROC (%) / FPR95 (%).

Method	NINCO	OpenImage-O	SSB-Hard	iImageNet-C	ImageNet-ES	iImageNet-R	ImageNet-V2	Average
<i>ImageNet-21k Pre-trained ViT-B/16</i>								
MSP	87.81 / 50.02	93.72 / 27.51	76.72 / 68.43	67.91 / 78.58	69.35 / 69.26	79.73 / 59.15	57.57 / 89.92	76.12 / 63.27
MLS	91.59 / 42.80	96.28 / 18.65	81.25 / 63.86	70.54 / 76.77	72.11 / 66.79	83.65 / 53.22	57.86 / 90.09	79.04 / 58.88
Energy	92.12 / 39.62	96.80 / 16.02	81.87 / 61.53	70.80 / 76.02	72.44 / 66.06	84.25 / 50.52	57.79 / 90.19	79.44 / 57.14
Mahalanobis	94.00 / 32.51	97.58 / 12.61	85.01 / 52.17	73.93 / 72.64	73.04 / 67.08	85.32 / 48.95	58.02 / 90.81	80.99 / 53.83
ViM	92.29 / 38.65	96.94 / 15.32	82.36 / 60.40	71.19 / 75.39	72.47 / 65.98	84.37 / 50.20	57.79 / 90.15	79.63 / 56.58
NECO	91.97 / 38.09	96.90 / 15.19	84.81 / 54.96	70.55 / 75.48	72.01 / 67.61	82.43 / 53.86	56.86 / 90.44	79.36 / 56.52
Procrustes-DS (Ours)	94.98 / 26.74	98.21 / 9.72	86.34 / 49.46	83.96 / 53.57	76.78 / 63.45	88.49 / 42.85	58.36 / 91.36	83.88 / 48.16